

HANDS-ON 9 [Advanced]

State Management – NgRx Store, Actions, Reducers, Effects & Selectors

Aim:

To implement NgRx state management in Angular by managing course and enrollment data using Store, Actions, Reducers, Effects, and Selectors.

Task 1: Set Up NgRx Store and Define Course State

Aim:

To configure the NgRx Store and manage course state using actions, reducers, and selectors.

Steps:

- Install NgRx packages and Redux DevTools.
- Configure StoreModule (or provideStore() for standalone Angular).
- Create Course Store files:
 - course.actions.ts
 - course.reducer.ts
 - course.selectors.ts
- Define actions:
 - loadCourses
 - loadCoursesSuccess
 - loadCoursesFailure
- Create reducer with:
 - courses
 - loading
 - error
- Register the reducer in the application.
- Create selectors:
 - selectCourseState
 - selectAllCourses
 - selectCoursesLoading
 - selectCoursesError
- Replace service calls in CourseListComponent with Store selectors.
- Dispatch loadCourses() in ngOnInit().

- Display course data using the async pipe.

Task 2: NgRx Effects for HTTP and Enrollment State

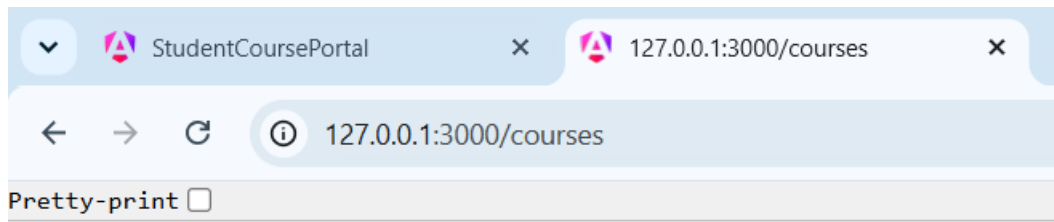
Aim

To handle asynchronous API requests using NgRx Effects and manage enrollment state using NgRx Store.

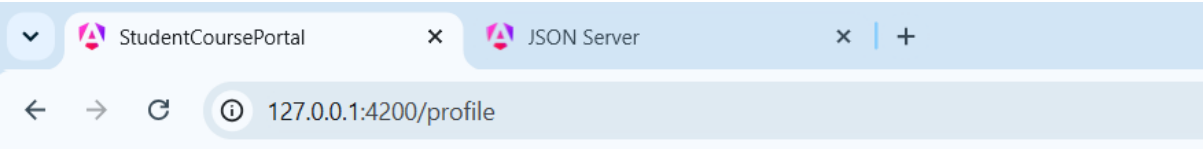
Steps

- Create `course.effects.ts`.
- Implement `loadCourses$` effect using:
 - `switchMap`
 - `map`
 - `catchError`
- Register `CourseEffects`.
- Verify the complete data flow:
 - Action → Effect → HTTP Call → Success Action → Reducer → Selector → Component.
- Create Enrollment Store:
 - `enrollment.actions.ts`
 - `enrollment.reducer.ts`
 - `enrollment.selectors.ts`
- Manage enrolled course IDs.
- Dispatch `enrollInCourse()` and `unenrollFromCourse()` actions.
- Use selectors to display **Enroll** or **Unenroll** button dynamically.
- Create cross-slice selectors to combine Course and Enrollment states.

Outputs:



```
[
  {
    "id": "1",
    "name": "Angular Fundamentals",
    "code": "ANG101",
    "credits": 4,
    "gradeStatus": "pending"
  },
  {
    "id": "2",
    "name": "RxJS and State Management",
    "code": "RXJ201",
    "credits": 3,
    "gradeStatus": "passed"
  },
  {
    "id": "3",
    "name": "API Integration Workshop",
    "code": "API301",
    "credits": 2,
    "gradeStatus": "pending"
  },
  {
    "name": "CNS",
    "code": "56",
    "credits": 5,
    "gradeStatus": "passed",
    "id": "Xt1a9hfflN8"
  },
  {
    "name": "SE",
    "code": "45",
    "credits": 5,
    "gradeStatus": "passed",
    "id": "PaAI3Gkd4Ww"
  },
  {
    "name": "New Course 8163",
    "code": "NEW8163",
    "credits": 3,
    "gradeStatus": "pending",
    "id": "K66oyBJo_RE"
  },
]
```



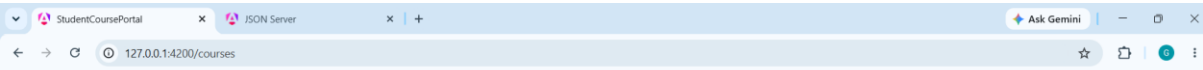
Student Course Portal

[Home](#)[Courses](#)[Enroll](#)[Profile](#)[Notification](#)

Student Profile

Enrolled Courses

Angular Fundamentals (ANG101)
RxJS and State Management (RXJ201)
SE (45)



Student Course Portal

[Home](#)[Courses](#)[Enroll](#)[Profile](#)[Notification](#)

Available Courses

Angular Fundamentals

ANG101

4

Pending

Update Course

Cancel

Search courses

Search

Angular Fundamentals Course Code: ANG101 Credits: 4 Status: pending Unenroll Edit Delete	RxJS and State Management Course Code: RXJ201 Credits: 3 Status: passed Unenroll Edit Delete	API Integration Workshop Course Code: API301 Credits: 2 Status: pending Enroll Edit Delete
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